$2\times\left(\frac{1}{2}\right)^4=\frac{1}{8}$. If *A* is *B*'s great-grandchild, the generation distance is 3 and the number of common 'ancestors' is 1 (*B* himself), so the relatedness is $1\times\left(\frac{1}{2}\right)^3=\frac{1}{8}$. Genetically speaking, your first cousin is equivalent to a great-grandchild. Similarly, you are just as likely to 'take after' your uncle ($\text{relatedness}=2\times\left(\frac{1}{2}\right)^3=\frac{1}{4}$) as after your grandfather ($\text{relatedness}=1\times\left(\frac{1}{2}\right)^2=\frac{1}{4}$).

For relationships as distant as third cousin $(2\times\left(\frac{1}{2}\right)^8=\frac{1}{128}$, we are getting down near the baseline probability that a particular gene possessed by *A* will be shared by any random individual taken from the population. A third cousin is not far from being equivalent to any old Tom, Dick, or Harry as far as an altruistic gene is concerned. A second cousin (relatedness = $\frac{1}{32}$) is only a little bit special; a first cousin somewhat more so ($\frac{1}{8}$). Full brothers and sisters, and parents and children are very special ($\frac{1}{2}$), and identical twins (relatedness = 1) just as special as oneself. Uncles and aunts, nephews and nieces, grandparents and grandchildren, and half brothers and half sisters, are intermediate with a relatedness of $\frac{1}{4}$.

Now we are in a position to talk about genes for kin-altruism much more precisely. A gene for suicidally saving five cousins would not become more numerous in the population, but a gene for saving five brothers or ten first cousins would. The minimum requirement for a suicidal altruistic gene to be successful is that it should save more than two siblings (or children or parents), or more than four half-siblings (or uncles, aunts, nephews, nieces, grandparents, grandchildren), or more than eight first cousins, etc. Such a gene, on average, tends to live on in the bodies of enough individuals saved by the altruist to compensate for the death of the altruist itself.